

**EXPERIMENT NO. 10****V CURVES OF SYNCHRONOUS MACHINE****AIM**

- To synchronize a 3 phase alternator to the supply mains using Dark lamp method
- Plot the V curves and inverted V curves when synchronous machine is acting as generator and motor at no load and constant power.

APPARATUS

Sl. No.	Name of Apparatus	Type	Range	Quantity
1	Voltmeter			
2	Voltmeter			
3	Voltmeter			
4	Ammeter			
5	Ammeter			
6	Rheostat			
7	Rheostat			
8	Tachometer			

PRINCIPLE

Assume constant power operation of a synchronous machine connected to an infinite bus. The equivalent circuit, neglecting the stator resistance, and the phasor diagram are shown in figures. For a 3 phase machine, the power transfer is $P = 3VI_a \cos \phi$. Because V is constant, for constant power operation $|I_a \cos \phi|$ is constant; that is, the in-phase component of the stator current on the axis of the phase V is constant. The locus of the stator current is therefore the vertical line passing through the current phasor for unity power factor.

In figure, phasor diagrams are drawn for three stator currents:

$$I_a = I_{a1}, \text{ lagging } V$$

$$= I_{a2}, \text{ lagging } V$$